LAWRANCE TAN WEI LUN

Graduate AI Engineer | LLM Applications · AI Automation · RAG

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Malaysian citizen · Coursework completed · Available to start immediately (full-time or internship)

SUMMARY

Final-year Computer Science & IT student (Massey University, GPA 3.76/4.0) who builds AI applications that a business can trust: document Q&A that cites its sources, and agents that automate routine work while handing risky decisions to people. Hands-on experience with real retail purchasing and inventory data. Fluent in English, Mandarin and Malay.

EXPERIENCE

Purchasing & Inventory Data Assistant – Yuen Sheng Trading, Masai, Johor Part-time · Ongoing

- Process supplier invoices and credit notes for 6+ suppliers (incl. Gardenia, Lam Soon, Nabati) into SmartAcc DynaMod, a client-server inventory system on SQL Server linked to AutoCount accounting.
- Worked out and documented DynaMod's Excel import rules (template format, price option, 12-digit barcode mapping, SST rounding via cent adjustment, credit-note knock-off), turning ad-hoc data entry into a repeatable process.
- **In progress:** building an AI-assisted entry tool for the shop (Python, Claude vision) that turns supplier invoice photos into DynaMod-ready import files, with automatic quantity, Sub Total and SST checks and protection against duplicate imports.

PROJECTS

Grounded RAG – Document Q&A with Citations | LangGraph, FastAPI github.com/weilun9/grounded-rag

- Built a Q&A service that answers only with cited sources and refuses when the documents do not contain the answer; a citation check after generation replaces any uncited answer with a refusal.
- Combined keyword and semantic search with a reranker; on a 45-question benchmark, top-1 retrieval accuracy rose from 85.7% to 91.4%.
- End to end on a local 9B model at zero API cost: 88.6% answer accuracy, 96.5% of citations from the correct source document, 10/10 out-of-scope questions refused, 3.2 s median latency.
- Shipped with a web UI, REST API, Docker image and GitHub Actions CI (27 unit tests plus automated retrieval evaluation).

AR Collections Agent – Invoice Follow-up Automation | Python, Claude github.com/weilun9/ar-collections-agent

- Built an agent (~4.5k lines, 63 tests) that ranks overdue invoices by risk and sends reminders via WhatsApp and Gmail, with alerts to the finance team on Slack; wrote the product spec first, including what the agent must not do.
- Kept customers safe: approved message templates only, a rule engine that can only tighten decisions, and a 48-hour window for account managers to take over before any customer contact.
- Limited the LLM to summarising case history, rejecting any summary containing numbers not found in the source records.

Serverless IoT Monitoring on AWS | AWS IoT Core, Lambda, DynamoDB, CloudFormation

- Built a Raspberry Pi-to-AWS telemetry pipeline over MQTT/TLS with a secured dashboard (OAuth 2.0 + PKCE), remote device control, CloudWatch alarms and 27 tests mocking AWS, fully defined in CloudFormation.

LLM Skill Evaluation – A/B Study | Prompt engineering, evaluation

- Benchmarked a custom LLM skill against a baseline on 142 graded items: accuracy rose from 90.1% to 98.6%; traced every remaining error and documented the limits of the test.

SKILLS

AI & LLM: LLM applications, AI agents, RAG, LangChain, LangGraph, prompt engineering, LLM evaluation, Claude Code, Ollama, OpenAI / Anthropic APIs, Hugging Face

Back end & cloud: Python, FastAPI, REST APIs, SQL, MongoDB, Docker, AWS (Lambda, IoT Core, DynamoDB, S3, Cognito, CloudFormation), GitHub Actions

Integration & tools: WhatsApp Cloud API, Gmail, Slack, Google Sheets, Git, Linux, pytest; familiar with SQL Server, DynaMod and AutoCount

Other languages: TypeScript / Next.js, C++, C# / ASP.NET MVC, Kotlin

EDUCATION

Bachelor of Information Sciences (Computer Science & Information Technology) Expected Mar 2027

Massey University, New Zealand – delivered at PSB Academy, Singapore · GPA 3.76/4.0 · Coursework completed · Massey "Excelling" recognition every semester

Relevant: Artificial Intelligence, Software Quality Assurance & Testing (A+), Object-Oriented Programming (A+), Algorithms & Data Structures (A), Database Development (A), IoT & Cloud Computing

Diploma in InfoComm Technology – PSB Academy, Singapore Apr 2024 – Dec 2024

LANGUAGES

English (fluent) · Mandarin (native) · Malay (fluent)